Spark Data Frame

1. Read
2. Processes
3. Write

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Spark Data Frame

Timestamp	Age	Gender	Country	state
2014-08-27 11:29:31	37	Female	United States	IL
2014-08-27 11:29:37	44	M	United States	IN
2014-08-27 11:29:44	32	Male	Canada	
2014-08-27 11:29:46	31	Male	United Kingdom	
2014-08-27 11:30:22	31	Male	United States	TX

Data Frame Schema

1. Column Names
2. Data Types

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File Edit View Navigate Code Refactor Run Tools VCS Window Help HelloSpark - utils.py - PyCharm

HelloSpark > lib > utils.py

Project

- HelloSpark C:\Users\prashant\Documents\GitHub\Spark-Programmir
  - app-logs
  - data
    - sample.csv
  - lib
    - __init__.py
    - logger.py
    - utils.py
  - HelloSpark.py
  - log4j.properties
  - spark.conf
- External Libraries
- Scratches and Consoles

HelloSpark.py | utils.py

```python
import configparser

from pyspark import SparkConf


def get_spark_app_config():
    spark_conf = SparkConf()
    config = configparser.ConfigParser()
    config.read("spark.conf")

    for (key, val) in config.items("SPARK_APP_CONFIGS"):
        spark_conf.set(key, val)
    return spark_conf


def load_survey_df(spark, data_file):
    return spark.read \
        .option("header", "true") \
        .option("inferSchema", "true") \
        .csv(data_file)
```

load_survey_df()

9: Git 6: TODO Terminal Python Console Event Log

PEP 8: W292 no newline at end of file 20:24 CRLF UTF-8 4 spaces Python 3.7 (HelloSpark) master

File Edit View Navigate Code Refactor Run Tools VCS Window Help HelloSpark - HelloSpark.py - PyCharm

HelloSpark > HelloSpark.py

Project

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HelloSpark.py | utils.py

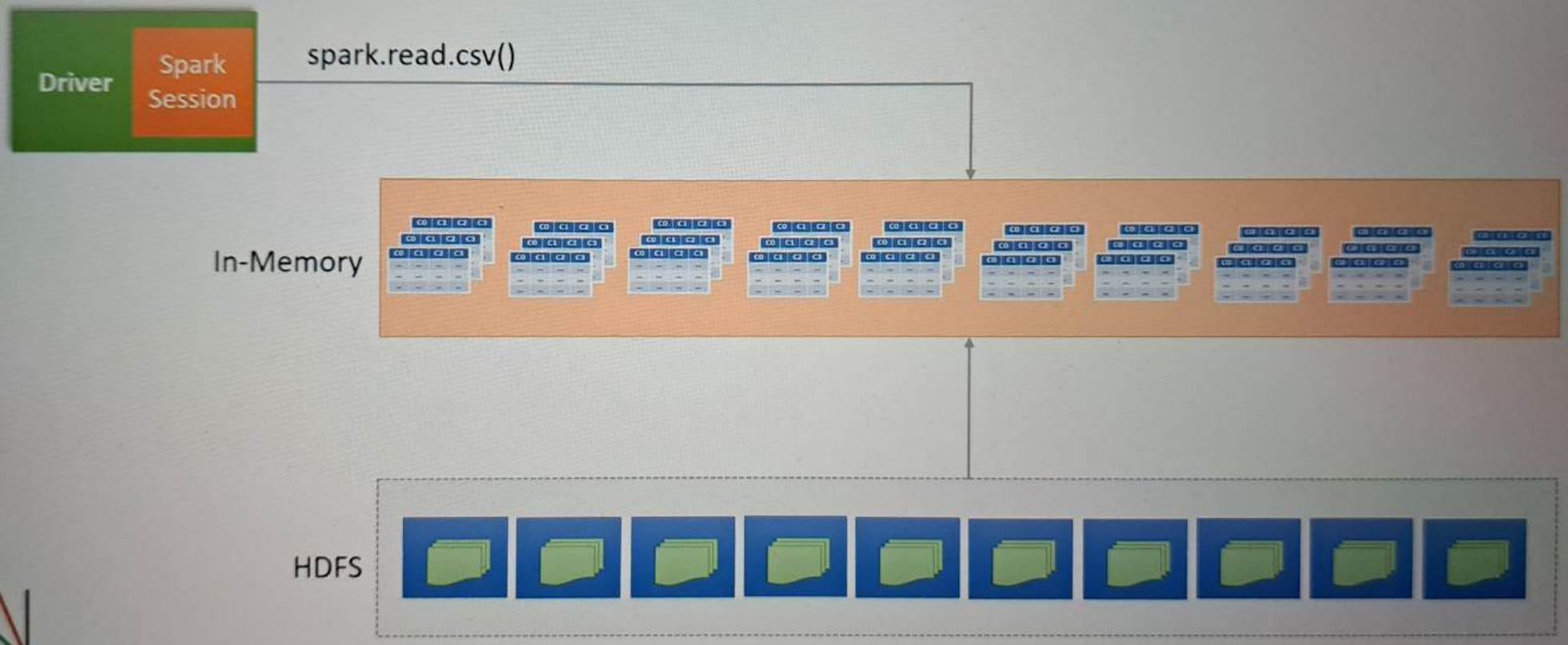
```python
import sys

from pyspark import SparkConf
from pyspark.sql import *

from lib.logger import Log4J
from lib.utils import get_spark_app_config, load_survey_df

if __name__ == "__main__":
    conf = get_spark_app_config()
    spark = SparkSession.builder \
        .config(conf=conf) \
        .getOrCreate()

    logger = Log4J(spark)

    if len(sys.argv) != 2:
        logger.error("Usage: HelloSpark <filename>")
        sys.exit(-1)

    logger.info("Starting HelloSpark")
    # Your processing code
    # conf_out = spark.sparkContext.getConf()
    # logger.info(conf_out.toDebugString())
    survey_df = load_survey_df(spark, sys.argv[1])
    survey_df.show()

    logger.info("Finished HelloSpark")
    # spark.stop()
```

if __name__ == "__main__"

9: Git | 6: TODO | 4: Run | Terminal | Python Console | Event Log

27:5 CRLF UTF-8 4 spaces Python 3.7 (HelloSpark) master

Spark Data Frame Partitions



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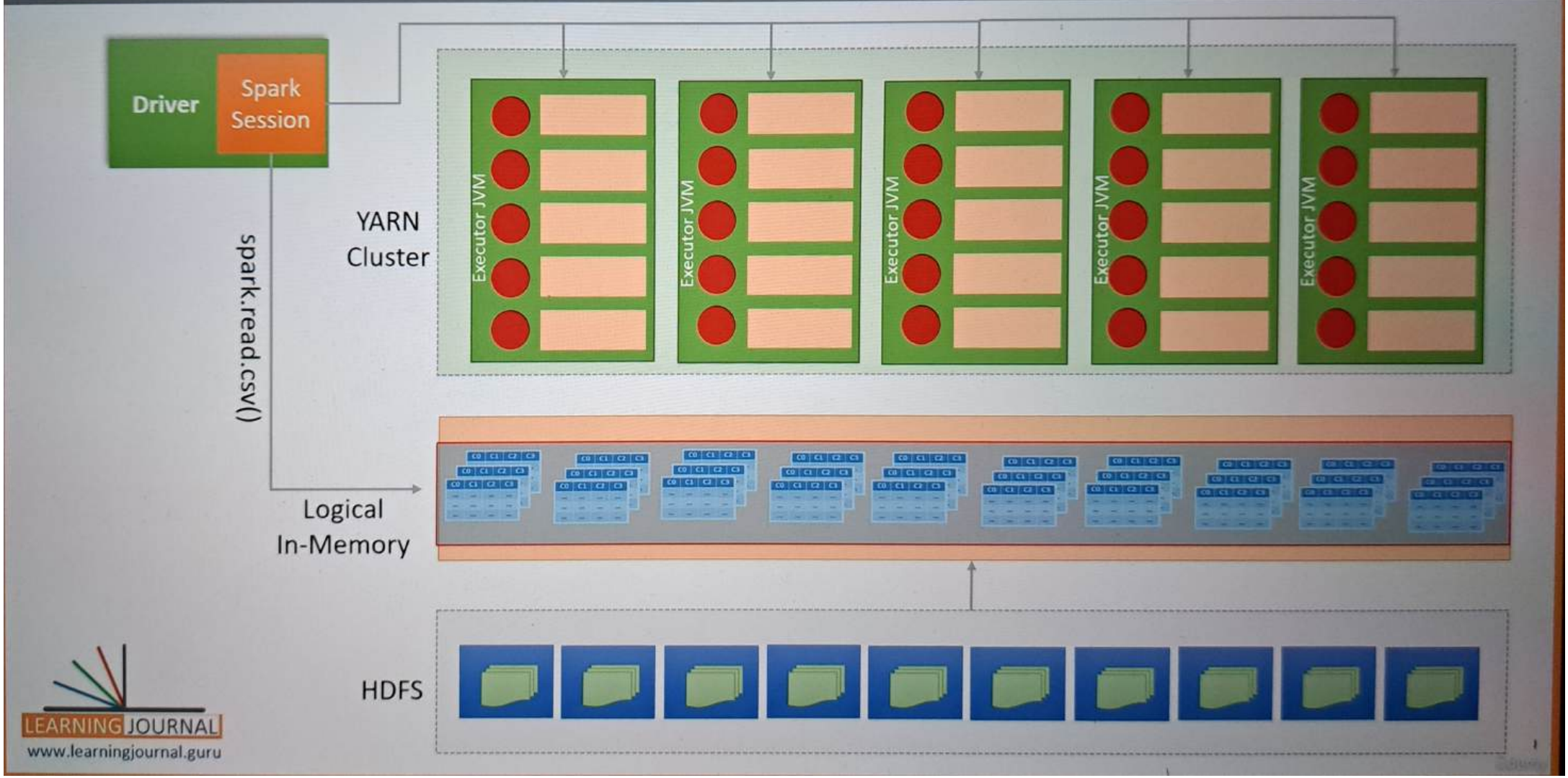
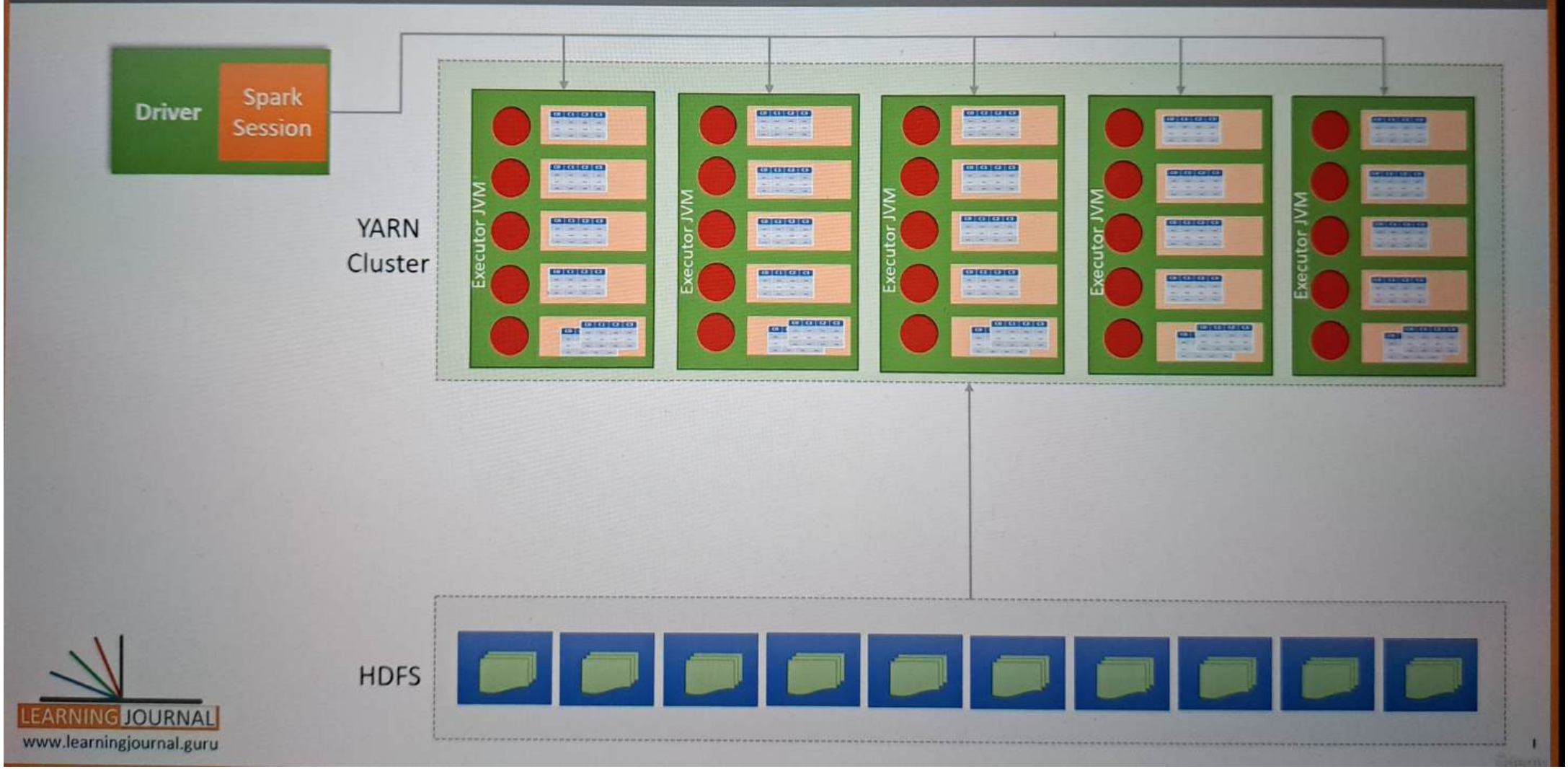
Spark Data Frame Partitions



Spark Data Frame Partitions